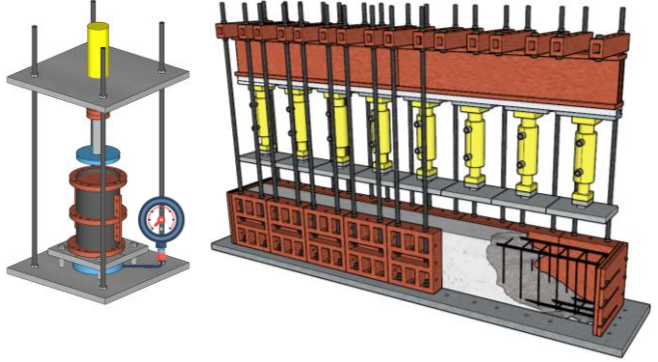
Compression cast concrete (CCC) is a type of concrete produced using the compression casting technique (CCT), where pressure is applied to consolidate the mix instead of traditional vibration methods. This approach enhances compressive strength, elastic modulus, density, and microstructure of concrete, while improving durability and reducing both production costs and CO_2 emissions



Predict properties of CCC

Design of CCC

Input

Cement (kg/m3)

Please input quantity in kg/m3

Sand (kg/m3)

Please input quantity in kg/m3

Gravel (kg/m3)

Please input quantity in kg/m3

Water (kg/m3)

Please input quantity in kg/m3

Casting pressure (MPa)

Please input value in MPa

Output

Compressive strength (MPa)

Density (kg/m3)

Elastic modulus (MPa)

Peak strain

Add a note
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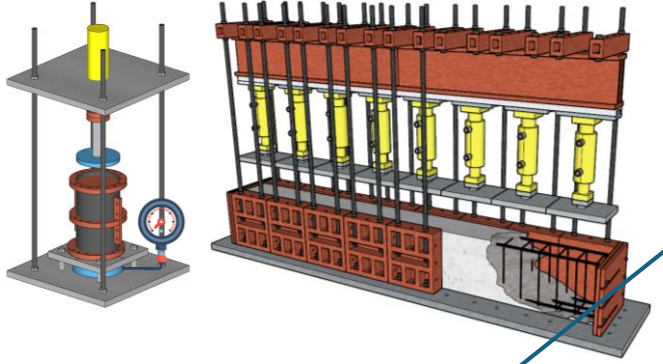
Compression cast concrete (CCC) is a type of concrete produced using the compression casting technique (CCT), where pressure is applied to consolidate the mix instead of traditional vibration methods. This approach enhances compressive strength, elastic modulus, density, and microstructure of concrete, while improving durability and reducing both production costs and CO_2 emissions



Predict properties of CCC

Design of CCC

Input

Target compressive strength (MPa)

Please input value from 20 to 60 MPa

Optimization objectives weight

(user preference based on economic, environmental, or performance perspective)

Weightage for cost (0 to 1)

Default value is 0.333

Weightage for embodied CO2 (0 to 1)

Default value is 0.333

Weightage for density (0 to 1)

Default value is 0.333

Optimal mix design

Cement (kg/m3)

Sand (kg/m3)

Gravel (kg/m3)

Water (kg/m3)

Casting pressure (MPa)

Cost and emission factors

(user defined based on project conditions)

Table 4 Cost and embodied carbon distances of concrete material

Material	p_{mi} (USD/kg)	c_{mi} (kg CO_2 /kg)	l_i (km)
Cement	0.0631	0.82	120.9
Sand	0.021	0.0036	63.4
CA	0.017	0.007	34.9
Water	0.000691	0.000181	0

Cost and embodied carbon of Optimal mix

Cost ($/m3)

Embodied CO2 (kg CO2/m3)